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Alphabet’s 2027 depreciation wave

What the AI build-out does to the 2027 P&L, and the cost guardrail that holds the operating margin

Nhi Nguyen

Public SEC data only. Not affiliated with Alphabet. E = estimate.

EXECUTIVE SUMMARY

A $53B depreciation year is mostly decided

$53B

2027E depreciation

Up 69% from $31B in 2026E; rises to 9.0% of revenue from 6.3%

83%

Already locked in

By assets bought through Sep 30, 2026. The 2027 CapEx budget moves it by under $1B

14%

Max growth in other costs

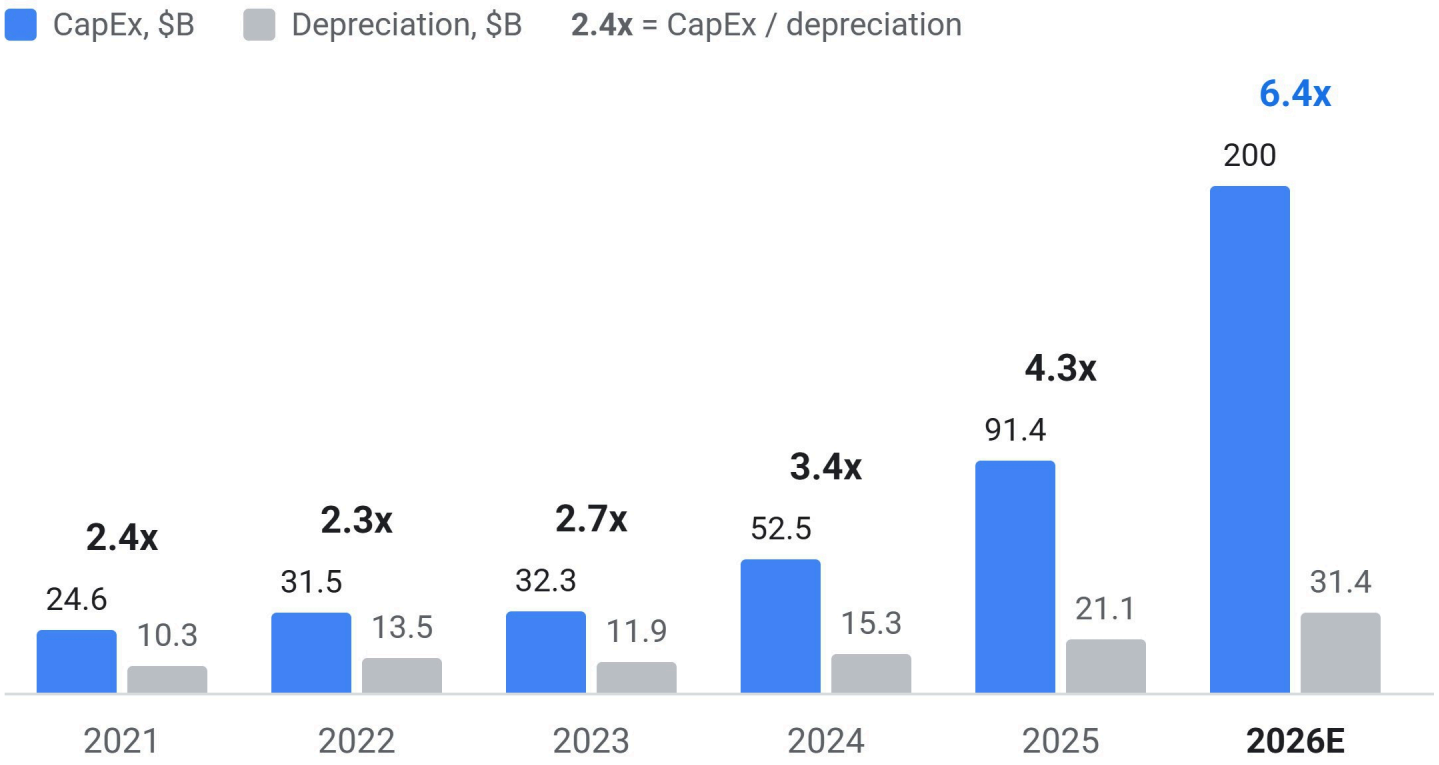
Holds the 2026E operating margin of 33.1%, versus 18.9% growth in 2026E

Recommendation: cap 2027 growth in costs other than depreciation at 14% in the plan, then re-test after the Q4 call, when 2027 CapEx guidance lands.

Source: Alphabet 10-K FY2025, 10-Q Q3 2026, Q3 2026 call; SEC XBRL; model estimates (E)

WHY IT IS HAPPENING

CapEx is running 6x ahead of depreciation



$145B

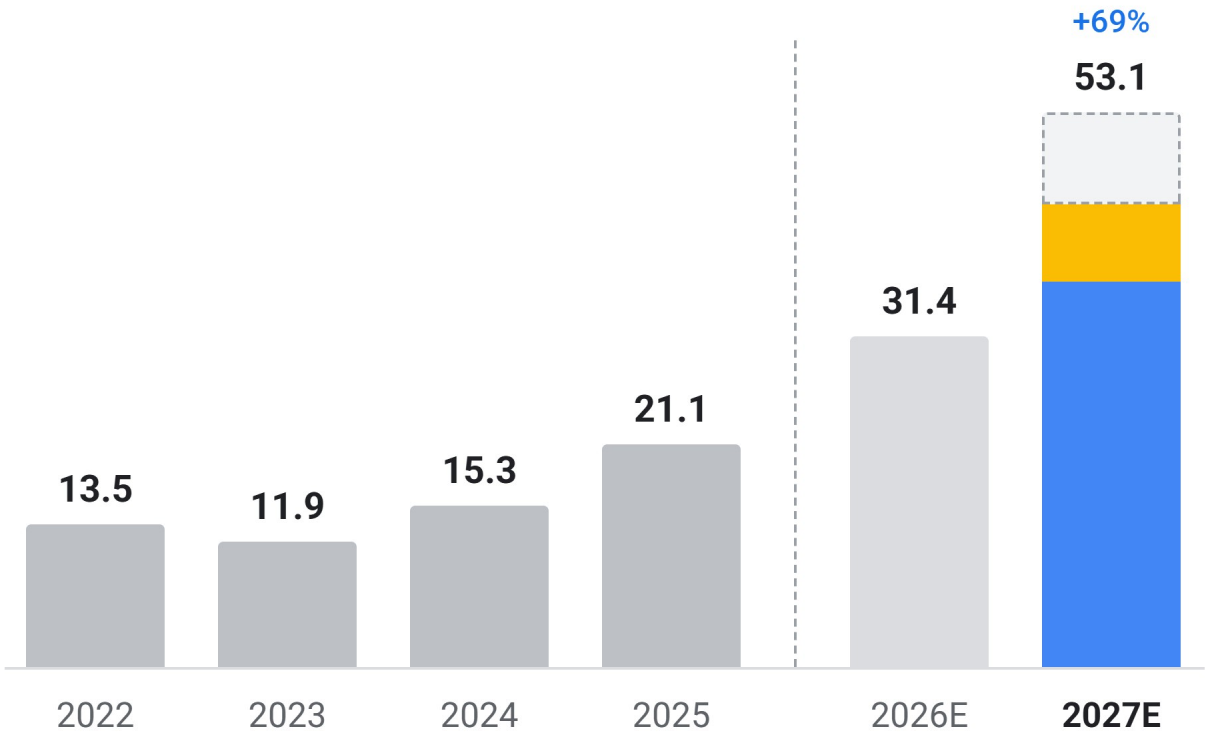
bought but not yet in service at Sep 30, 2026, up from $79B in December. It depreciates once it goes live.

Source: SEC XBRL (10-K, 10-Q). 2026E: CapEx at guidance midpoint; depreciation = 9M + model Q4

2027 FORECAST

2027E depreciation: $53B, 83% already locked in

Depreciation, $B. 2022-2025 reported; 2026E-2027E projected



What makes up 2027E's $53.1B

$36.6B
run-off of assets already in service

$7.3B
construction in progress going live

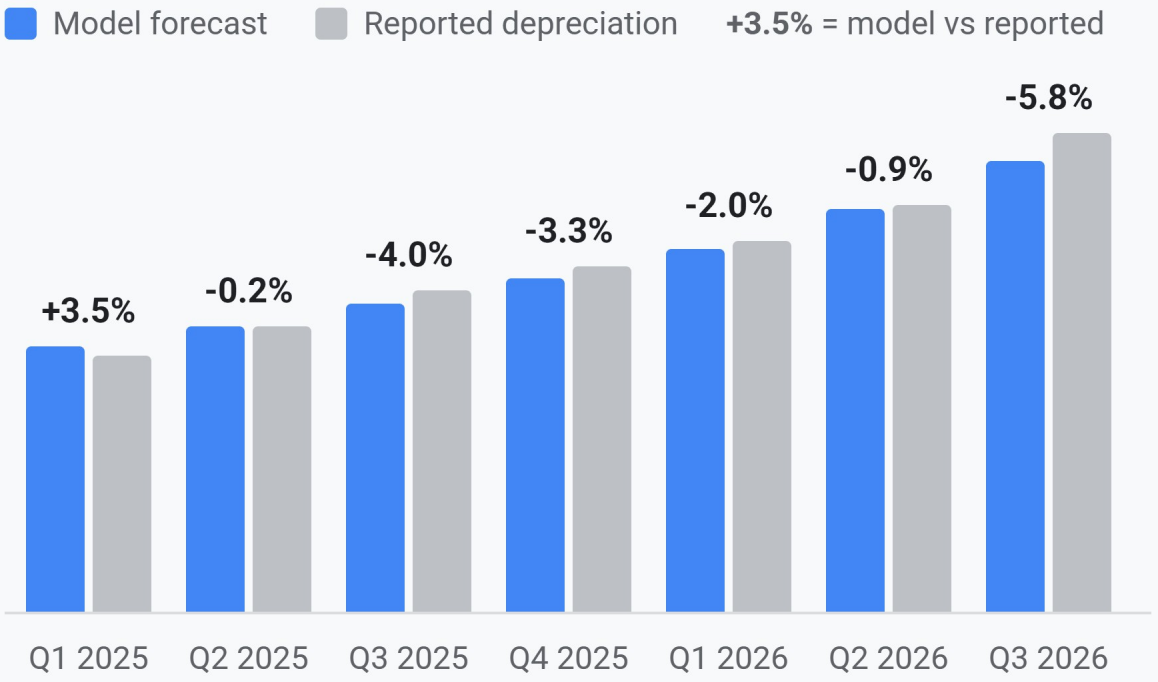
$9.3B
depends on spend from Q4 2026 on

Locked in: $43.9B, or 83%. A $250B or $350B 2027 CapEx budget moves 2027E by under $1B.

Source: SEC XBRL (10-K, 10-Q); 2026E-2027E from the vintage model, base case 2027 CapEx $300B

CAN WE TRUST IT?

Tested out of sample: 2.8% average miss



2.8%

average miss per quarter, 7 quarters out of sample

- Fit on 2021-2024 data only, then run on reported CapEx
- FY2025 within 1.3%; worst quarter Q3 2026 at -5.8%
- Misses grew in 2026 as spend tilted to faster-depreciating servers; with the CFO's 65% server mix, the full fit puts Q3 2026 within 1.4%

Source: SEC XBRL quarterly depreciation; vintage model fit on data through Dec 2024

MARGIN IMPACT

To hold a 33% margin, other costs grow 14%

33.1%
2026E operating margin

→

-2.7
points: depreciation climbs to 9.0% of revenue, from 6.3%

→

+0.1
points: other costs grow 18.9%, in step with 19% revenue growth

→

30.5%
2027E margin if other costs keep 2026's pace

GUARDRAIL

Grow other costs no more than 13.7% and the 2027E margin holds at 33.1%

Each extra point of revenue growth allows about 1.1 points more cost growth.

2027 revenue growth	Max other-cost growth	Margin at 2026 pace
+15%	9.3%	28.0%
+19% (base)	13.7%	30.5%
+23%	18.2%	32.7%

2026E = 9M reported + Q4 at revenue +22%, other costs +20% YoY. Other costs = all but depreciation

SCENARIOS AND RISKS

CapEx moves 2028; useful life moves 2027

2027 CapEx	2027E depreciation	2028E depreciation
$250B (+25%)	$52.3B	$79.6B
$300B base (+50%)	$53.1B	$84.6B
$350B (+75%)	$54.0B	$89.6B

Spread across cases: $1.7B in 2027E, $10.1B in 2028E.

Every case shares the same $43.9B locked-in base. New 2027 spend goes live through the year and depreciates mostly from 2028.

BIGGER 2027 RISK: USEFUL LIFE

+$13.7B

2027E depreciation if server and network lives drop from 6 to 5 years in January 2027

About 2.3 points of 2027 margin. Amazon cut a subset of servers from 6 to 5 years in January 2025, citing the pace of AI.

Server share of spend 5 points higher or lower: about $1.2B more or less 2027E depreciation.

Source: vintage model scenarios; Amazon change from its 10-K as summarized by Hudson Labs

RECOMMENDATION

Plan 2027 for the depreciation already bought

1

Cap 2027 growth in costs other than depreciation at 14%

Versus 18.9% in 2026E. Holds the operating margin near 33% at 19% revenue growth.

2

Show locked-in depreciation next to CapEx in monthly reporting

So every CapEx approval shows what it adds to 2027 and 2028 depreciation before it is signed.

3

Re-test the plan after the Q4 2026 call

Triggers: 2027 CapEx guidance outside $250-350B, any useful-life change, or Q4 2026 depreciation outside $9.0-9.9B (model: $9.4B).

Guardrail assumes 19% revenue growth; each extra point of growth allows about 1.1 points more cost growth

APPENDIX

How the model works

1 · Data

SEC XBRL API

269 facts: CapEx, depreciation, revenue, operating income, 2007 to Q3 2026

2 · Clean

SQL pipeline

BigQuery dialect, tested on DuckDB: latest filing wins, quarters from YTD, 5 tie-out checks

3 · Model

Python vintage engine

Spend split by asset type, go-live lag, straight-line lives; fit and backtested

4 · Plan

Excel, memo, deck

Live scenarios and margin guardrail; 15 checks; ties to Python at default inputs

Key assumptions

- 2026 CapEx $200B (guidance midpoint); 2027 base $300B
- Servers and network: 65% of 2026+ spend, 6-year life
- Q4 2026: revenue +22%, other costs +20%; 2027 revenue +19%

Sources

- Alphabet 10-Q Q3 2026 and 10-K FY2025
- Q3 2026 earnings call transcript
- SEC XBRL API (Nov 8, 2026); Webull; Hudson Labs

Portfolio exercise on public filings only; not affiliated with Alphabet. E = estimate